Master's Thesis

Can AI data centres solve the energy flexibility problem?



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Key findings at a glance

AI data centres can help with the energy flexibility problem, offering several terawatt-hours of geographical flexibility each year. By moving computation instead of electricity, operators can align demand with favourable renewable conditions.

489

hyperscale facilities mapped from open data, with 19.4 GW of estimated capacity

4.18 TWh/yr

shiftable when work may change place and hour (2.30% of demand)

2.52 TWh/yr

shiftable by changing location only, in the same hour (1.39% of demand)

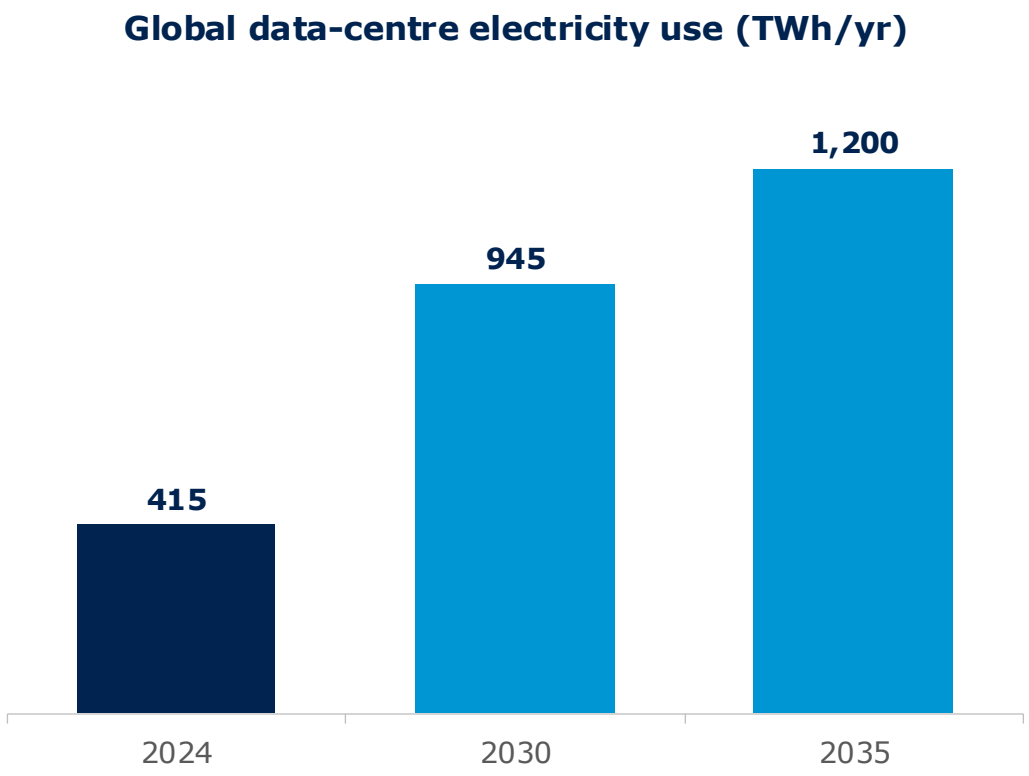
12.95 TWh/yr

daily shifting by 2035 in the accelerated IEA growth scenario

The limit is spare capacity at the receiving sites, not the share of flexible work. Even with the hour fixed, 60% of the daily potential remains, because distant regions offer better conditions at the same moment.

Reference case: 85% utilisation, 25% flexible share, six candidate hours, current and planned sites.

Rising demand meets variable supply



Data centres used about 1.5% of global electricity in 2024. Hyperscalers already run 37% of all data-centre capacity.

Demand is rising

The IEA expects data-centre electricity use to almost triple by 2035, mainly because of AI.

Supply is variable

Solar output follows daylight and wind follows the weather. When demand cannot adapt, surplus generation is curtailed.

Workloads can move

AI training, batch jobs and some inference can wait or run elsewhere. Their electricity demand can follow better conditions.

Source: International Energy Agency (2025), Energy and AI. Base Case projections.

Move computation, not electricity

Data centre A

Weak wind and sun in this hour.
It sends flexible work away.

Flexible workload

It moves over data networks that already exist.

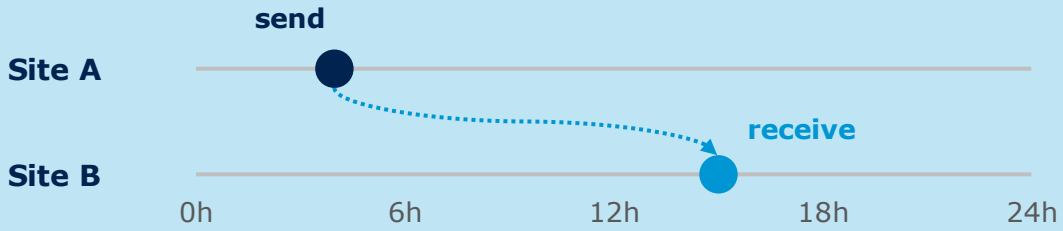
Its electricity demand moves with it.

Data centre B

Strong wind and sun in this hour.
It receives the work if it has room.

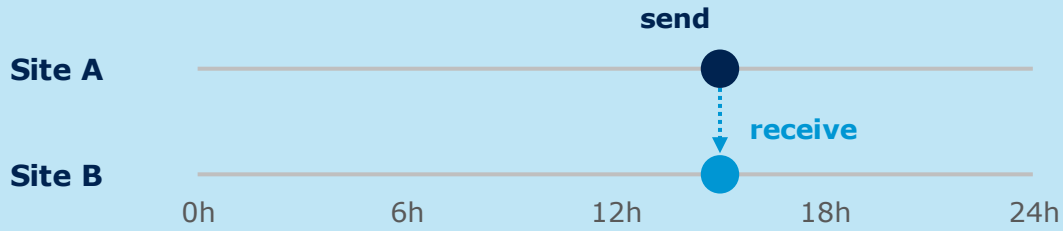
Daily mode: place and hour can change

Work may run at another site and at another hour of the same day.



Same-hour mode: only the place changes

Work runs at another site but in the same hour.



Source: Zhang et al. (2020) describe such transfers as virtual links between grids.

Research question

MAIN QUESTION

To what extent can the flexible share of electricity demand in hyperscale data centres be shifted across major global regions over the next decade?

SUB-QUESTION 1 – DEMAND

What is the electricity demand per data-centre cluster for hyperscale data centres in the next decade?

SUB-QUESTION 2 – SHIFTABILITY

To what extent is this demand flexible for geographical load shifting, given capacity and timing constraints?

SUB-QUESTION 3 – RENEWABLES

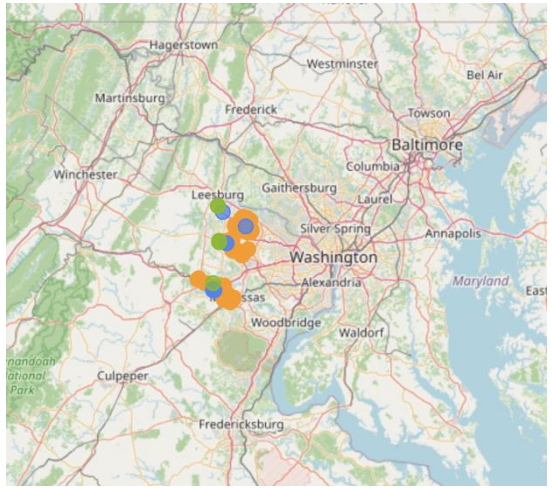
How could this flexibility help data centres align their load with variable renewable energy availability?

Approach in five steps

Facility database	Power capacity	Hourly inputs	Shifting model	Projections
Amazon, Google and Microsoft data centres from OpenStreetMap and Overture Maps	The floor area of each building is turned into electrical capacity	Wind and solar availability and a measured load profile, 2023 to 2025	An optimisation moves flexible demand within each operator's own network	Results scaled with the IEA growth scenarios for the sector
489 **facilities in 29 countries**	**19.4 GW** **today, 24.2 GW with planned sites**	**87** **clusters with hourly data**	**4.18 TWh/yr** **of demand can move each year**	**3** **growth scenarios to 2035**

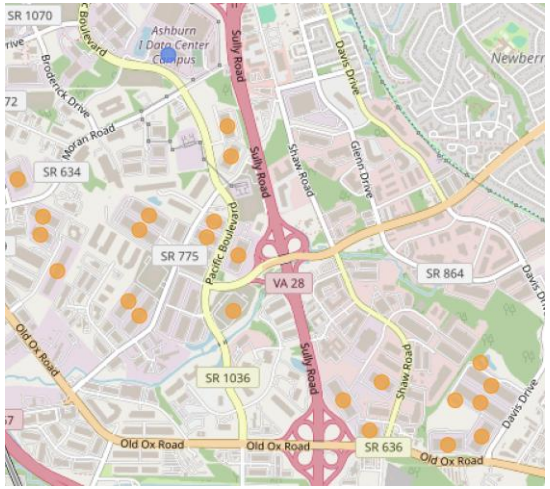
Data: OpenStreetMap, Overture Maps, Renewables.ninja, UK Power Networks, IEA (2025).

Tracing each facility to a building



REGION

Data centres in Northern Virginia



CAMPUS AREA

Facility locations around Ashburn



BUILDINGS

Markers beside Amazon buildings



ONE BUILDING

Mapped outline of Amazon IAD57

Matching records to buildings

Outlines are used directly. For a site boundary, the buildings inside are added up. For a point, buildings within 150 m count.

Counting the floors

Floors come from OpenStreetMap or Overture Maps, or from building height divided by 6 m. Without evidence, one floor is assumed.

Map data © OpenStreetMap contributors.

From floor area to megawatts

1

Gross floor area

The building footprint multiplied by the number of floors

2

IT capacity

60% of the floor area holds IT equipment, at 3,000 W per m²

3

Facility capacity

IT capacity multiplied by the operator's PUE, which adds cooling and other overhead

WORKED EXAMPLE GOOGLE, THORNTON (COLORADO)

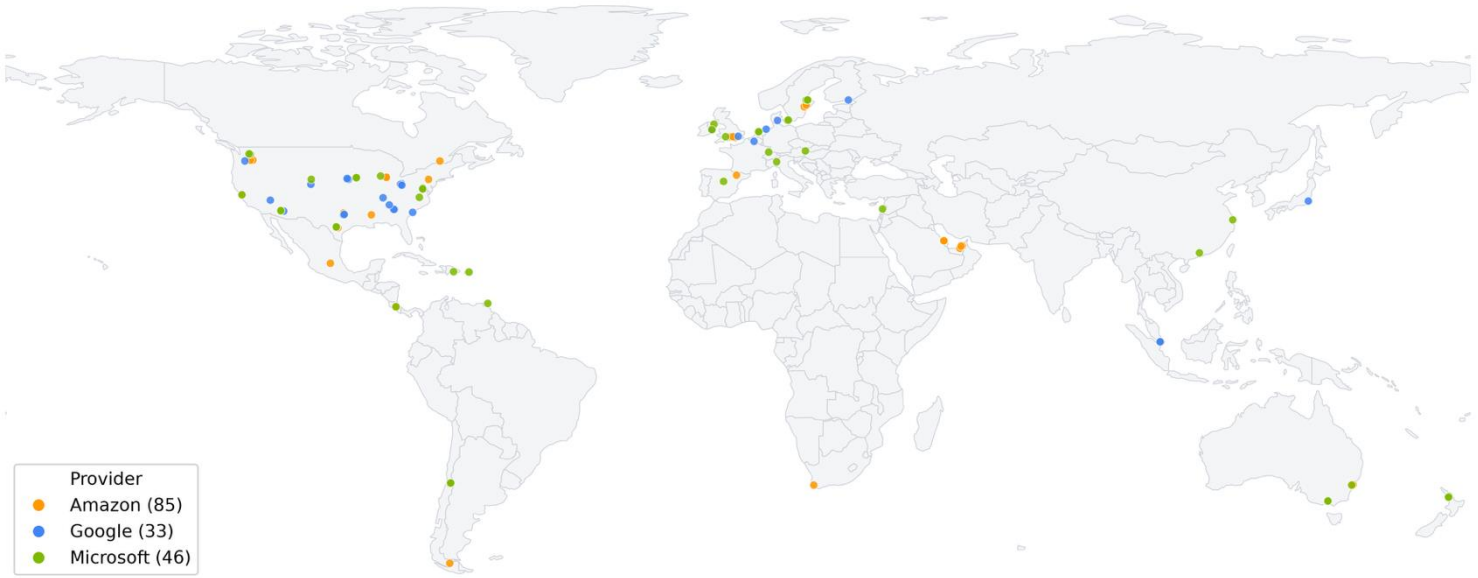
7,995 m²	15,990 m²	28.78 MW	31.37 MW
mapped footprint	gross floor area with 2 recorded floors	IT capacity	facility capacity with Google's PUE of 1.09

Operator PUE

Google **1.09**	Amazon **1.15**	Microsoft **1.17**

Source: Nicholson (2024) for density benchmark. Operator sustainability reports for PUE values.

19.4 GW, mostly in North America



14.54 GW	3.39 GW	0.98 GW	0.38 GW	0.13 GW
North America	Europe	Asia Pacific	Latin America	Middle East & Africa

SANITY CHECK, UNITED STATES

14.36 GW

Bottom-up estimate of this thesis for the three operators

15.59 GW

Top-down IEA estimate for all US hyperscalers

The bottom-up figure is 7.8% lower. The IEA figure also covers other hyperscale operators.

From buildings to clusters

489

Facilities

Single data-centre buildings or sites, each with its own capacity estimate.

202

Campuses

Facilities of one operator within 1 km of each other. 164 today plus 38 planned locations.

87

Clusters

Campuses within 100 km in the same country and time zone. They share one wind and solar profile.

Each operator keeps its own network

A cluster holds the capacity of every operator separately. Work only moves between sites of the same company.

Two network layers

The model runs for today's 164 campuses and again with 38 planned locations, each sized at 25 to 200 MW from similar campuses.

When to send and when to receive

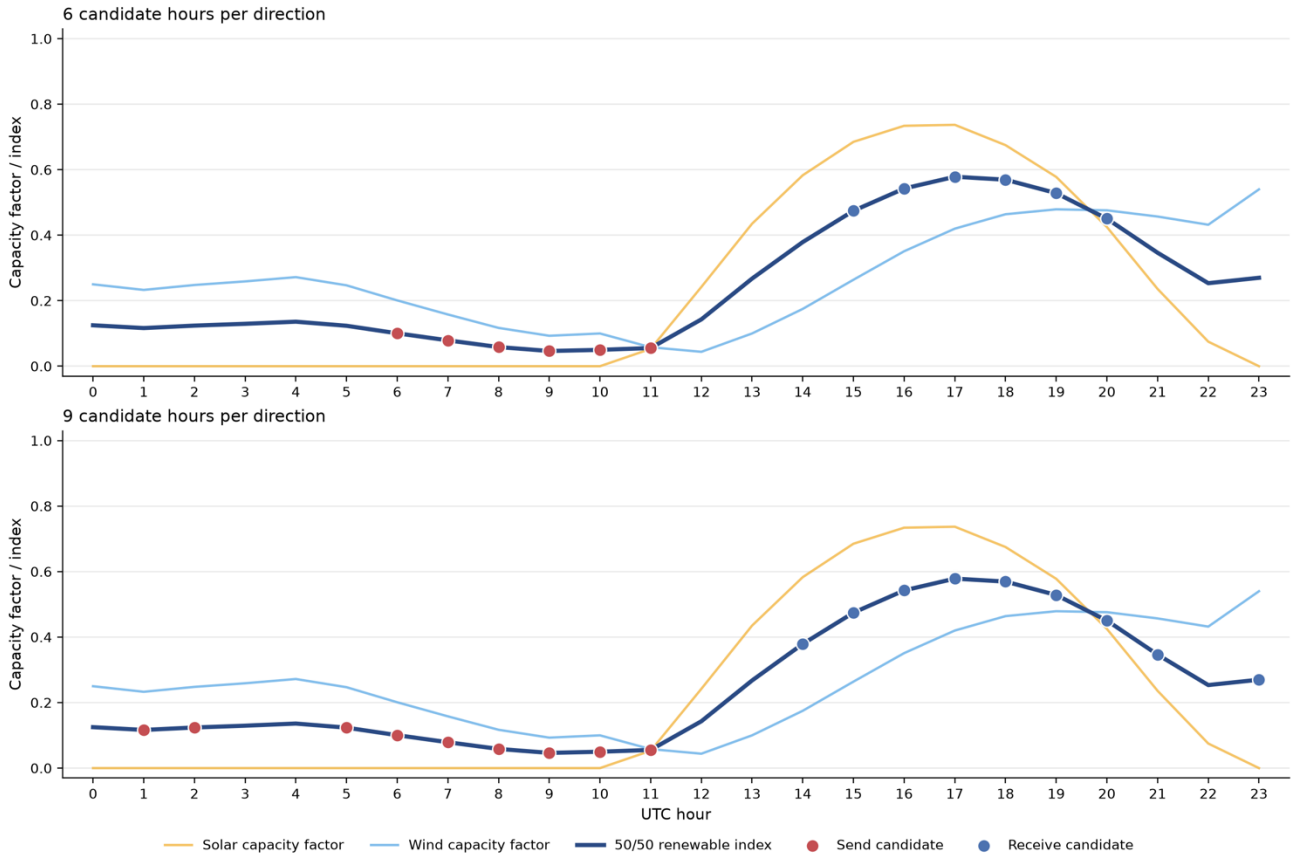
WIND AND SOLAR AVAILABILITY

$$R = 0.5 \cdot CF_{solar} + 0.5 \cdot CF_{wind}$$

CF is the capacity factor: the share of a plant's maximum output. R runs from 0 to 1.

1. Hourly solar and wind capacity factors for all 87 clusters, 2023 to 2025.
2. Each day, the six hours with the least wind and sun become send hours. The six with the most become receive hours.
3. Work only moves if the receive hour has higher availability than the send hour.

Example: Northern Virginia on 5 September 2025, hours in UTC.



Source: Renewables.ninja, based on MERRA-2 reanalysis (Pfenninger and Staffell, 2016).

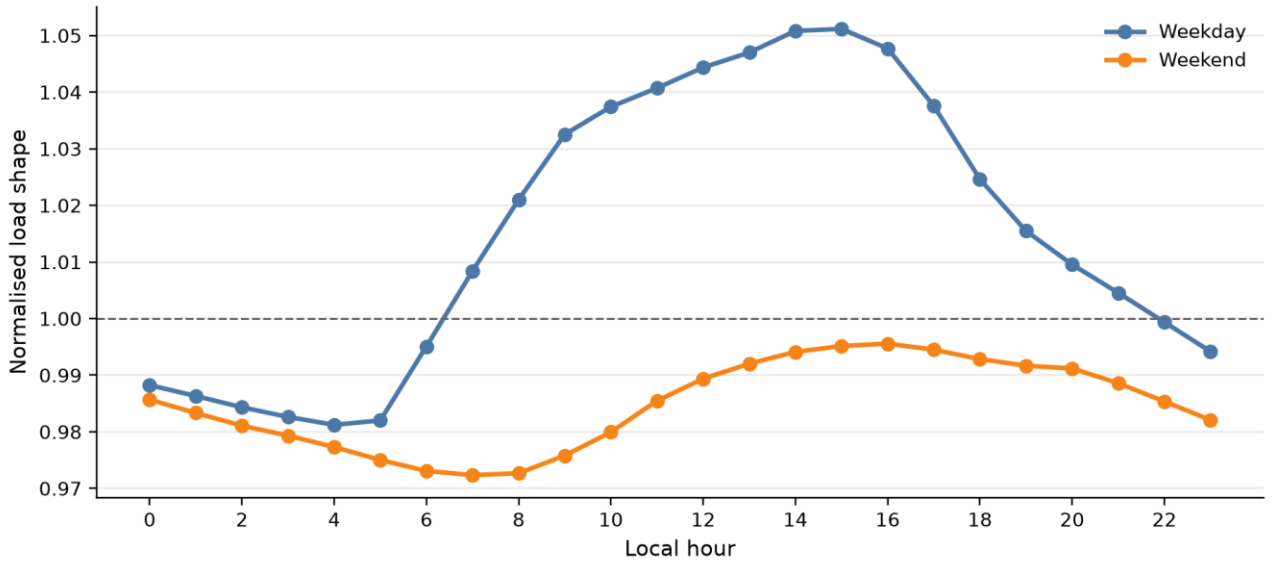
Demand is nearly flat over the day

LOAD OF A SITE IN HOUR t

$$L = min(C, C \cdot u \cdot q)$$

- C Facility capacity from the floor-area estimate
- u Utilisation of 85% in the reference case, with 80% and 90% tested
- q Hourly shape from the profile, applied in each site's local time

Normalised profile of 91 UK data centres, 2023 to 2025.



The profile only describes how demand changes over time. Utilisation sets the level.

Source: UK Power Networks (2026), Data Centre Demand Profiles.

Sites can send more than others receive

SENDING BUDGET

$B = f \cdot L$

f is the flexible share of 25%, with 20% and 30% tested

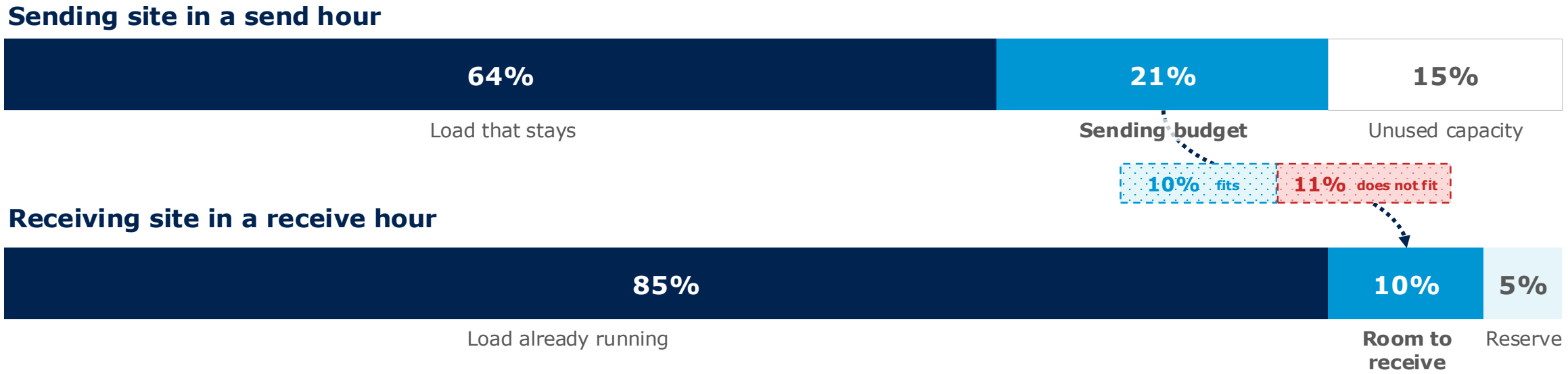
ROOM TO RECEIVE

$H = max(0, 0.95 \cdot C - L)$

Never negative. 5% of capacity always stays free.

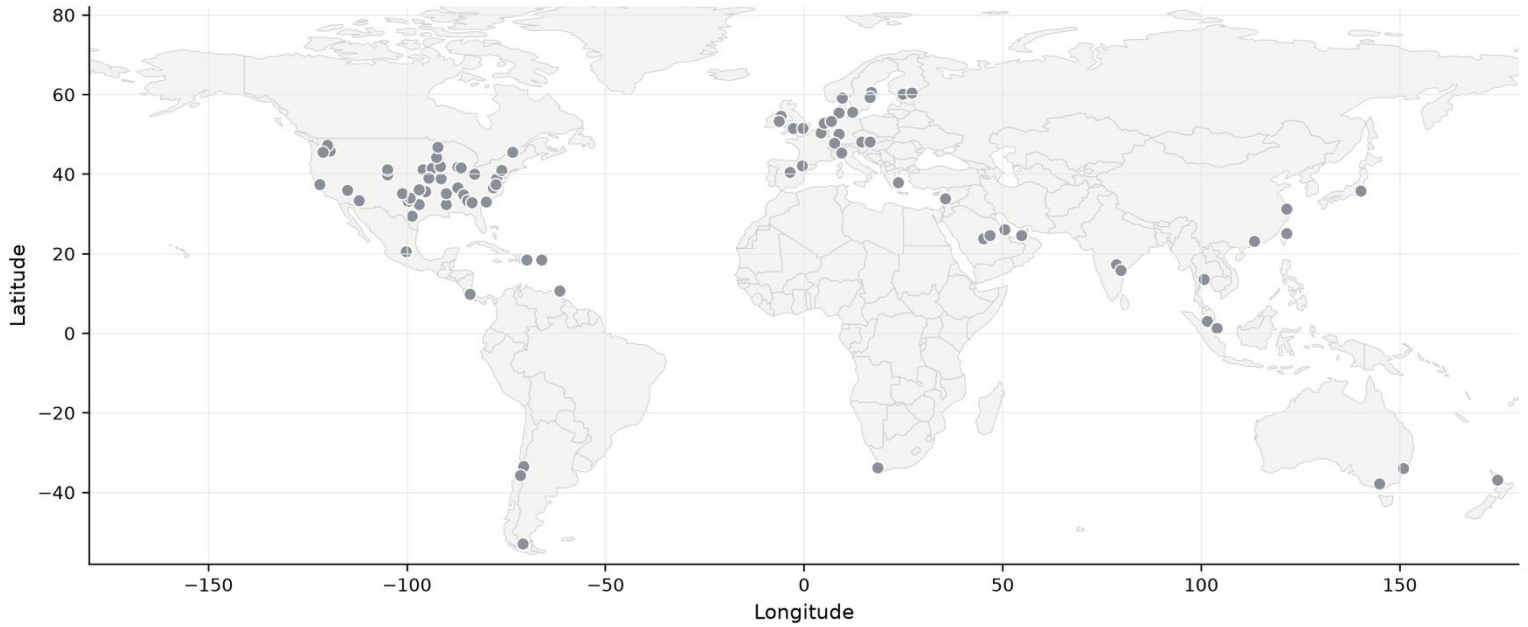
A site can give away 21% of its capacity, but another can only take 10%.

Illustration at 85% utilisation and average load. Each bar shows 100% of a site's capacity.



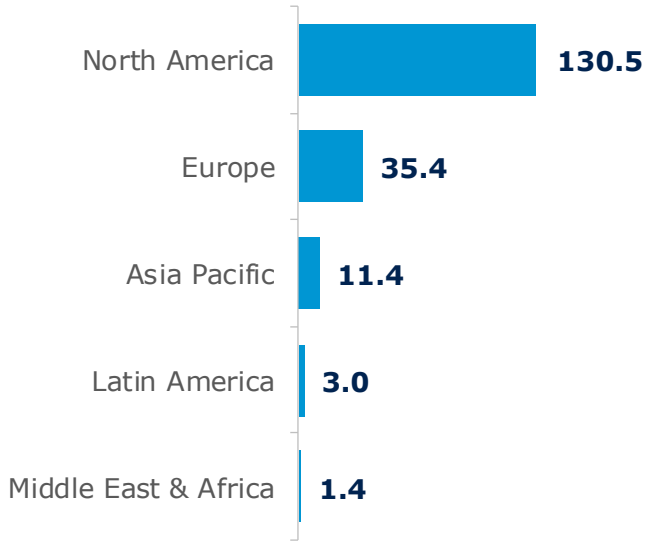
Work only moves within one operator, to a different cluster, and towards higher wind and solar availability.

182 TWh a year across 87 clusters



Each point is one cluster of nearby campuses that share one wind and solar profile.

Annual demand by region (TWh)

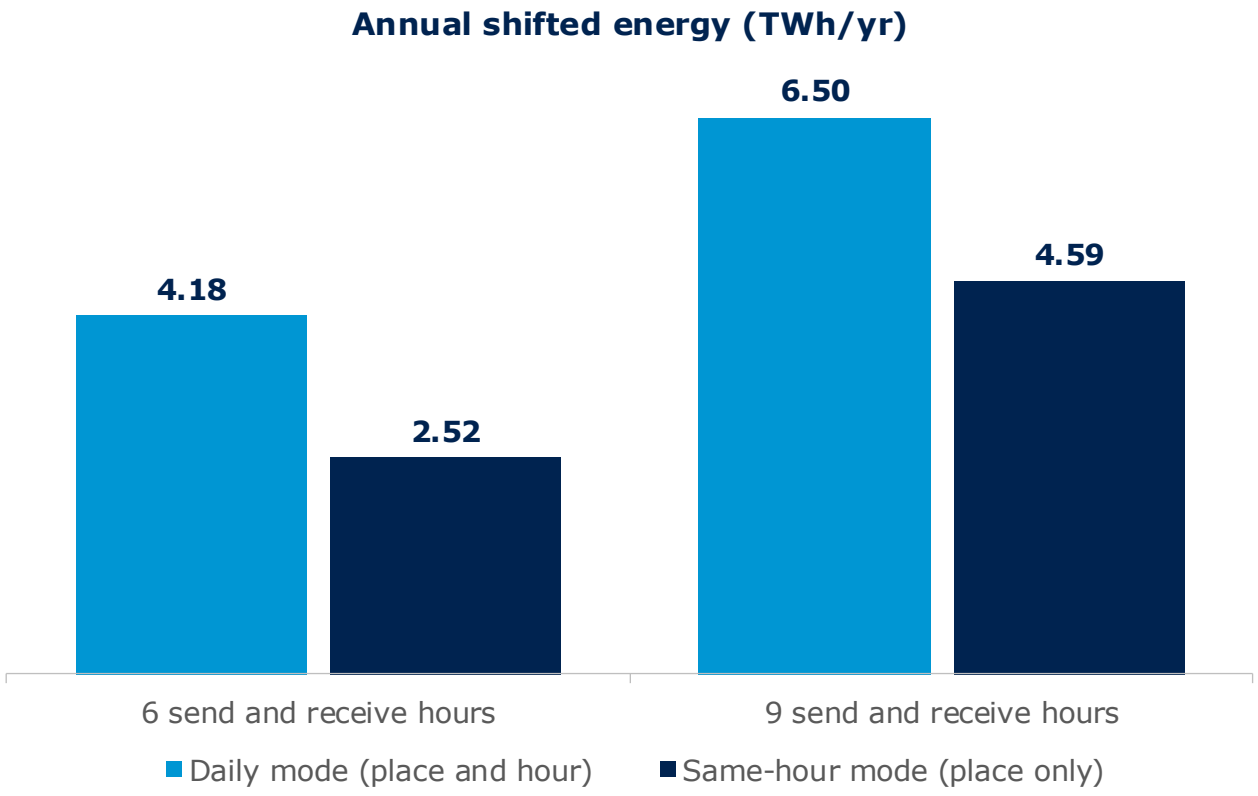


24.22 GW
network capacity including the 38 planned locations, which add 24.7% to demand

1.02 TWh/yr
median demand of one cluster

Three operators, including planned sites, at 85% utilisation. All data centres used about 415 TWh in 2024.

2.3% of demand can move each year



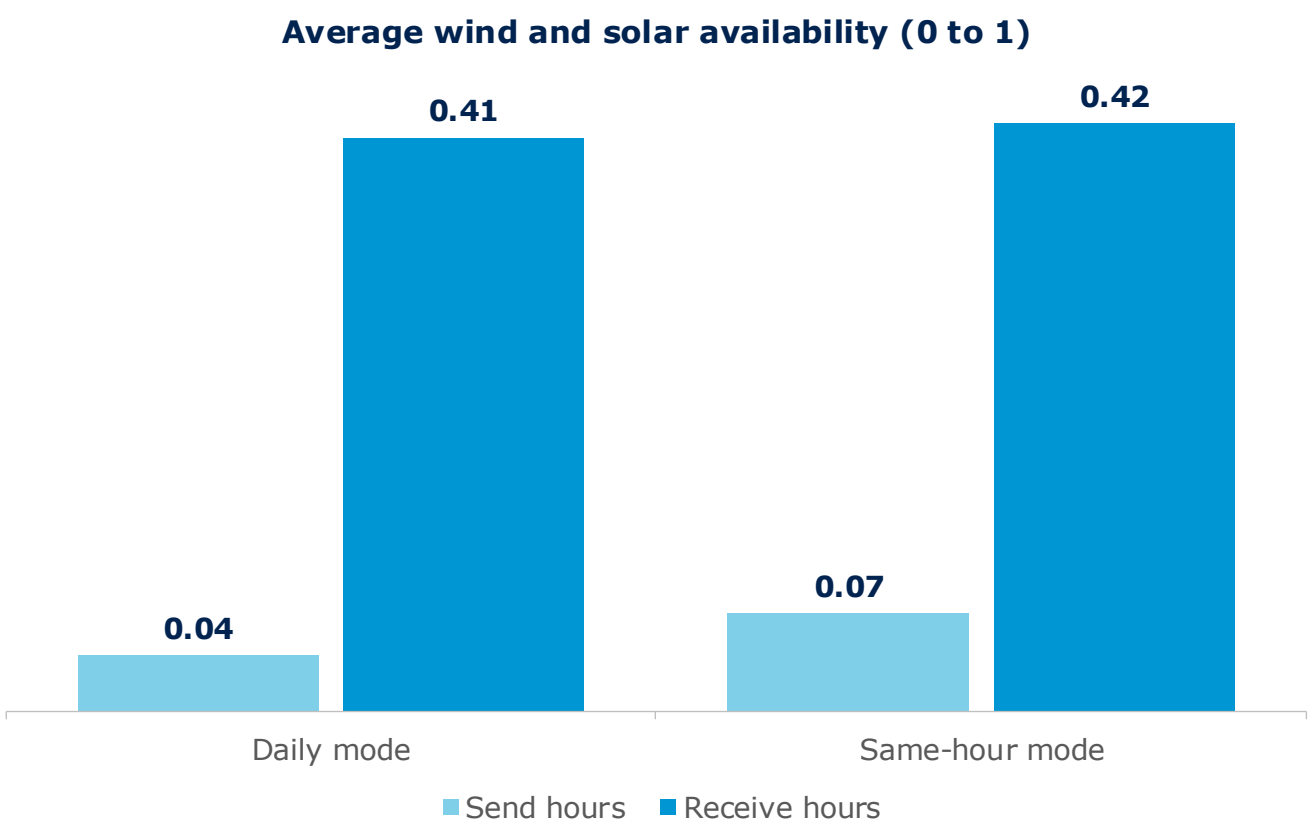
2.3% of demand moves in daily mode, and 1.4% in same-hour mode

60% of the daily volume remains when the hour is fixed. With nine hours, it is 71%.

0.6% difference between the weather years 2023 to 2025.

Reference case: 85% utilisation, 25% flexible share, current and planned sites.

Demand moves to much better hours



10 times higher availability in the receive hours than in the send hours, in daily mode

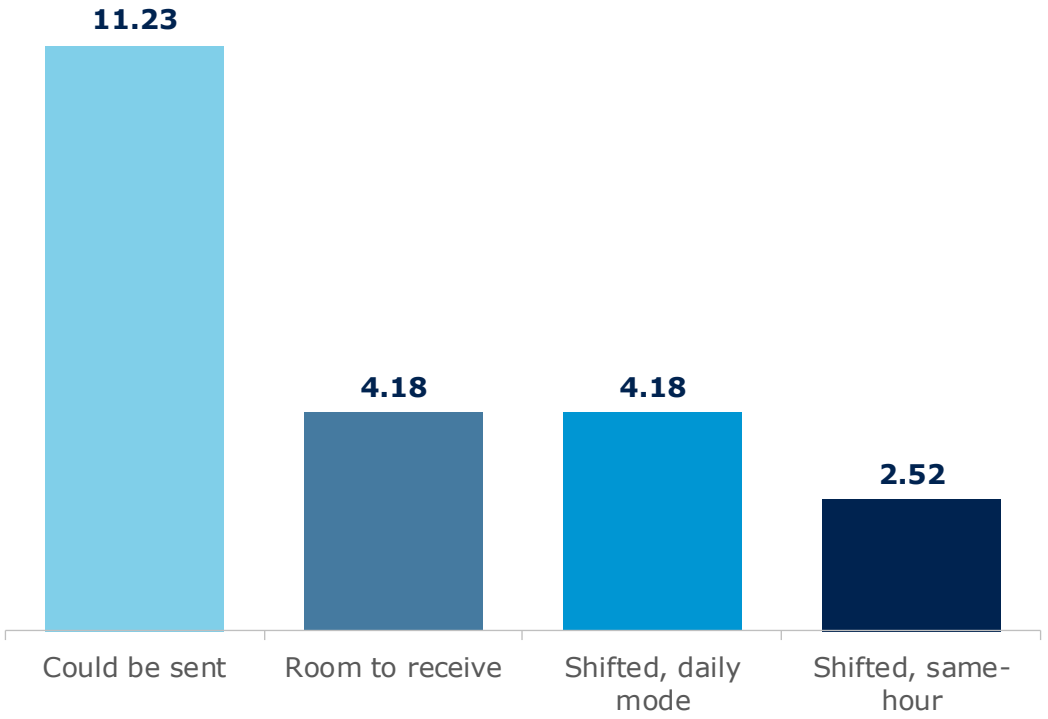
6 times higher availability in same-hour mode, reached by changing location alone

5.6 h average time shift in daily mode

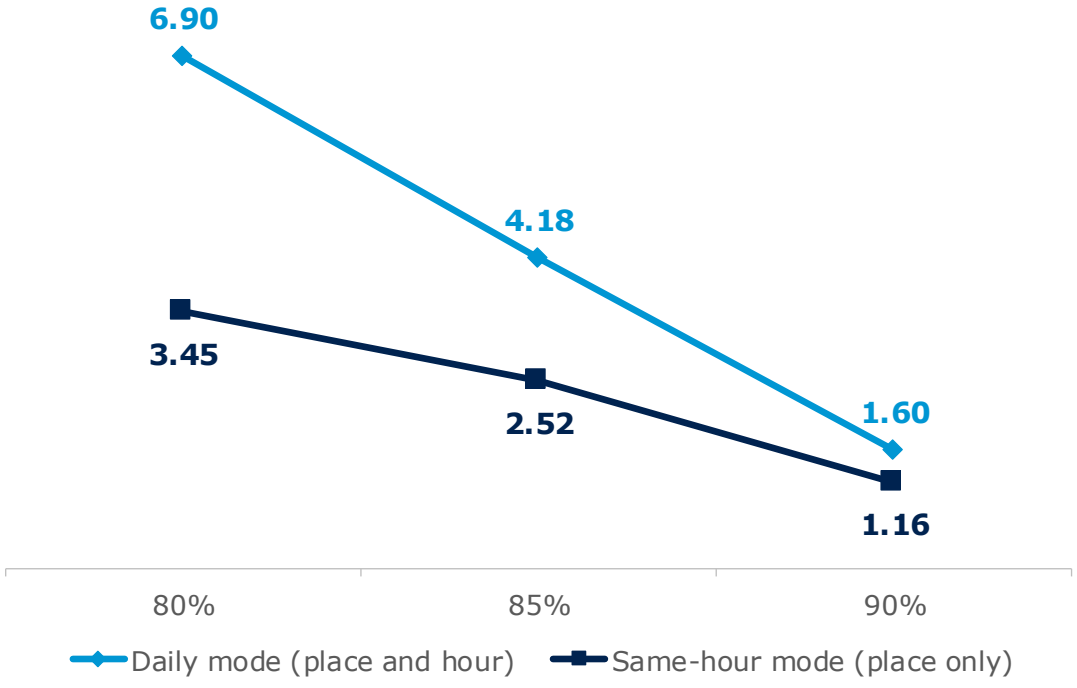
Availability runs from 0 (no wind or sun) to 1. It describes resources, not delivered renewable power or emissions.

Receiving capacity is the bottleneck

More could be sent than received (TWh/yr)



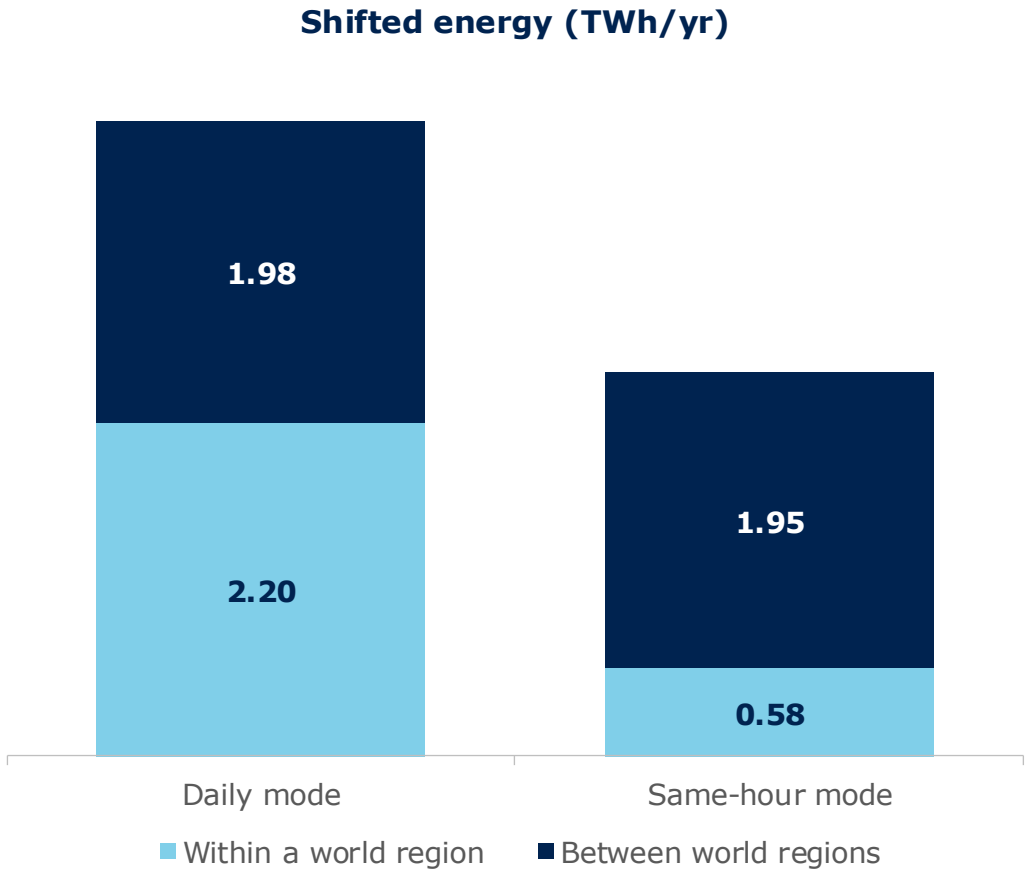
Higher utilisation leaves less room (TWh/yr)



A larger flexible share does not help. From 20% to 30%, daily shifting stays at 4.18 TWh per year.

Reference case: 85% utilisation, 25% flexible share, 6 send and receive hours per day, current and planned sites.

Geography does the heavy lifting



47%

of daily-mode shifting crosses world regions

77%

of same-hour shifting crosses world regions

With the hour fixed, most transfers within a region disappear. Transfers between regions stay almost unchanged at about 1.95 TWh per year.

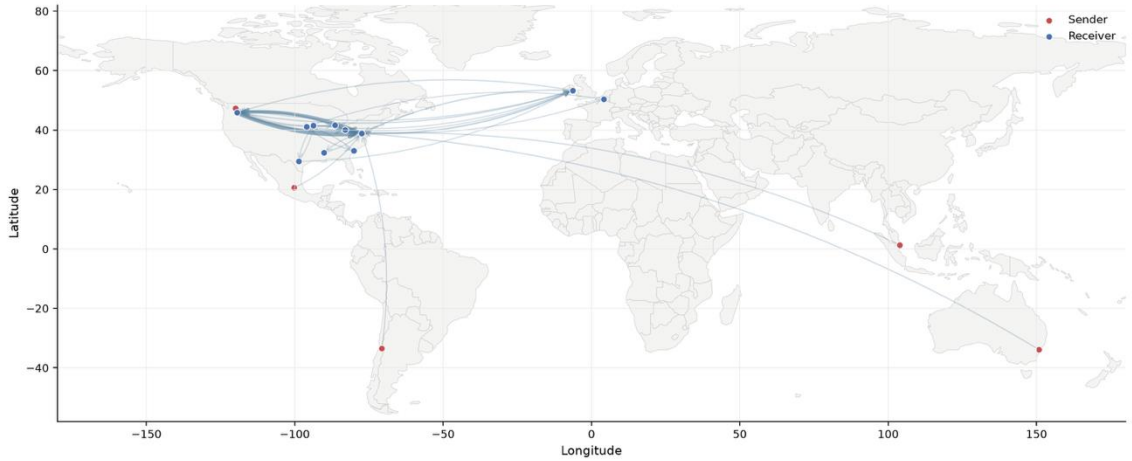
A distant region has different daylight and weather at the same moment.

Reference case: 85% utilisation, 25% flexible share, 6 send and receive hours per day, current and planned sites.

Same-hour flows cross the oceans

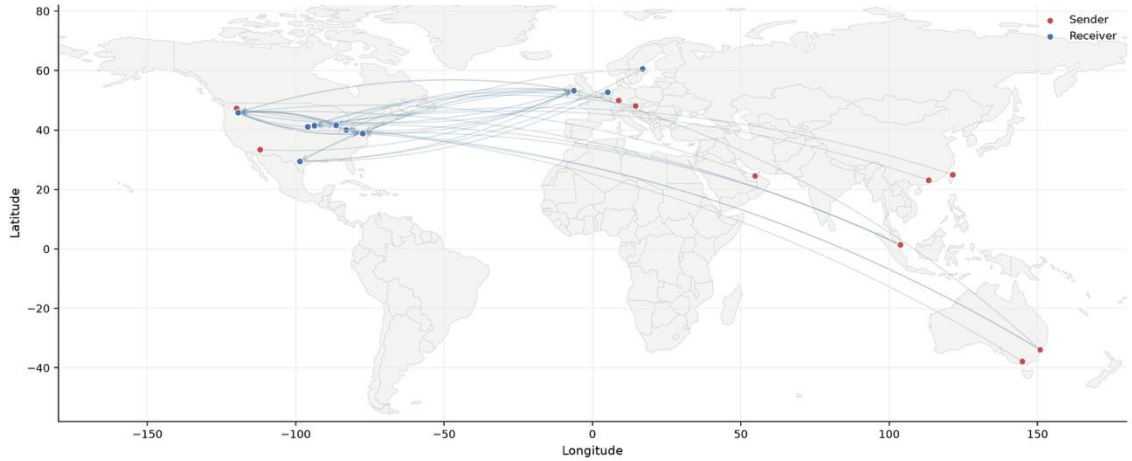
Daily mode (place and hour)

Most of the largest links connect clusters within North America.



Same-hour mode (place only)

More links cross the oceans to Europe, the Middle East and Asia Pacific.



2.09 TWh/yr

move within North America in daily mode, half of all shifted energy

255 GWh/yr

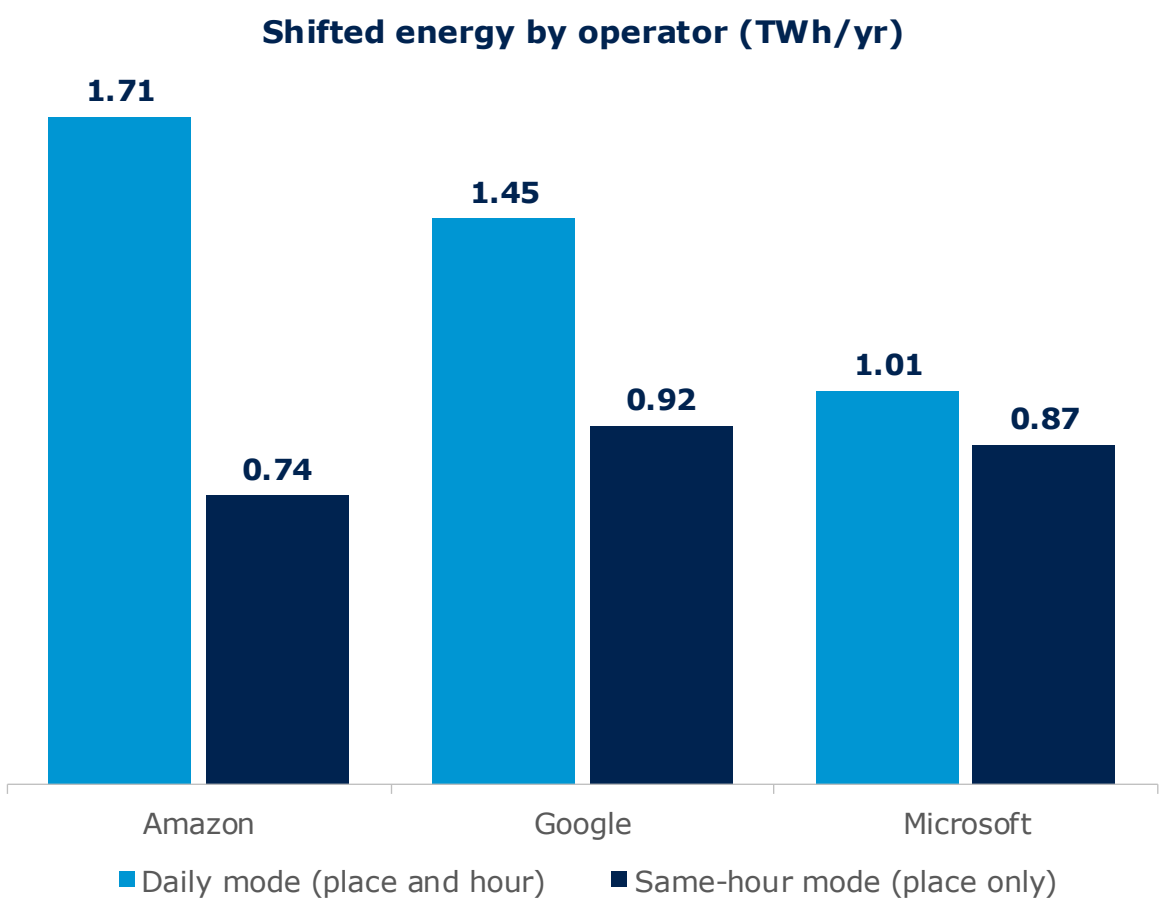
flow on the largest link, from eastern Oregon to Northern Virginia

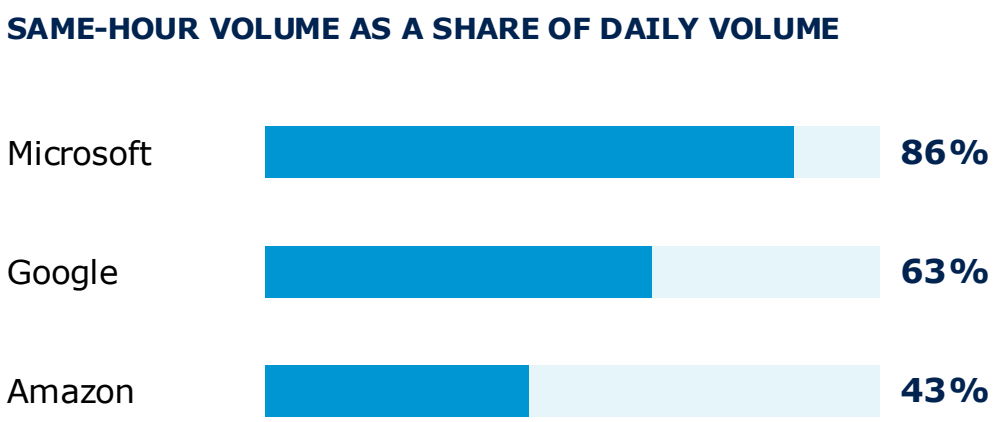
0.60 TWh/yr

flow from Europe to North America, the largest flow in same-hour mode

The 35 largest connections. Line width shows annual shifted energy. Red marks senders, blue marks receivers.

Network reach shapes the potential



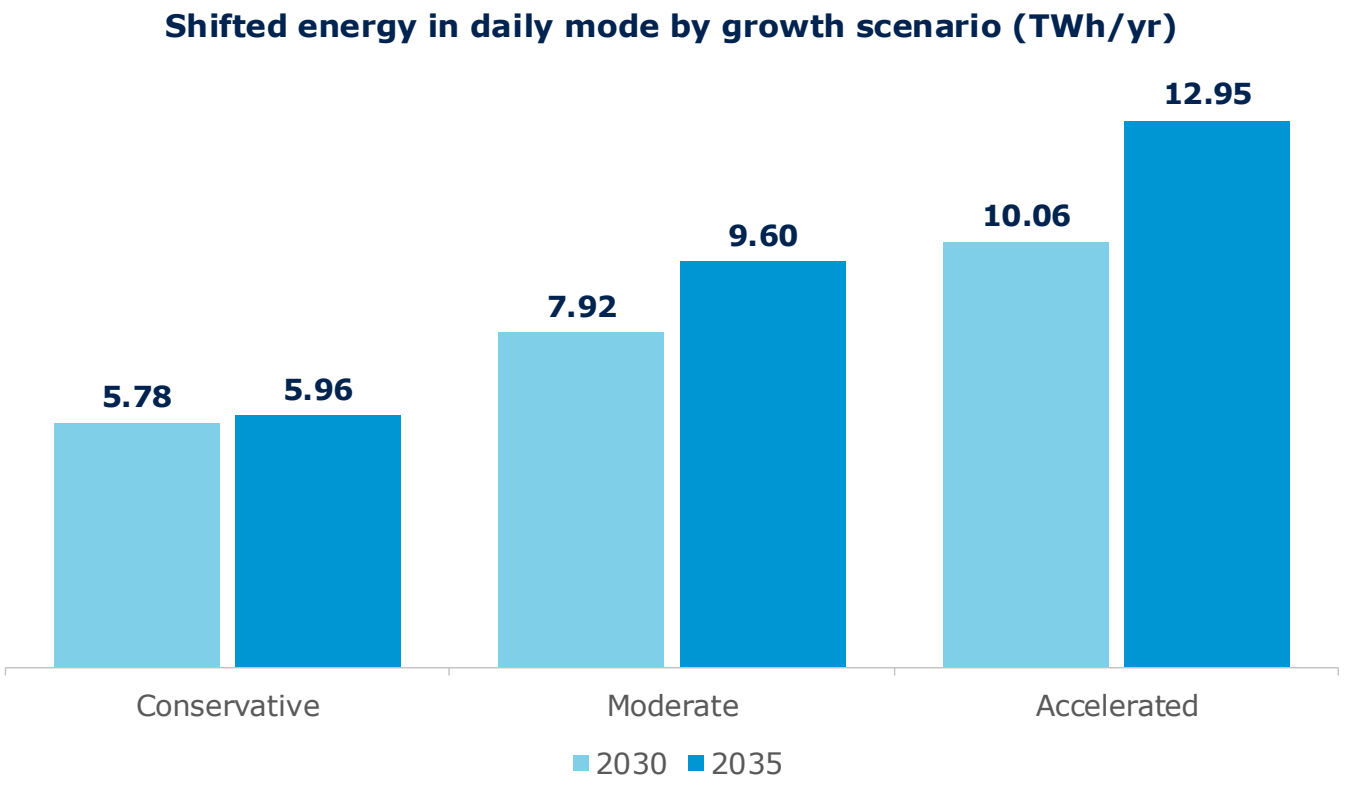


In daily mode, all three operators shift about 2.3% of their demand.

With the hour fixed, Amazon keeps only 43%, although it has the largest demand. How far a network spreads matters more than how flexible its work is.

Reference case: 85% utilisation, 25% flexible share, 6 send and receive hours per day, current and planned sites.

Outlook: up to 13 TWh a year by 2035



2.3%

of demand stays movable in every scenario, because demand and room to receive grow together

3.60–7.82 TWh/yr

can move in same-hour mode by 2035

Scenarios follow the IEA. Global hyperscale capacity grows from 36 GW in 2024 to 64, 103 or 139 GW in 2035.

Median weather year, 6 send and receive hours per day. With 9 hours, daily mode reaches 9.27 to 20.13 TWh/yr in 2035.

What this means for operators

Plan flexibility across the whole network

A site with excellent wind and solar can receive little work if its own load already fills its capacity. Workload placement and utilisation targets need to be set together. What a location adds depends on the network around it.

Coordinate the scheduling

A simpler greedy rule moves almost the same energy as the optimisation. The optimisation reaches better hours with shorter time shifts. Its value lies in where and when the work runs, not in how much of it moves.

Give flexibility a reward

Whether operators shift work depends on the reward. A market could pay data centres for shifting demand according to price differences across locations and times. This gives them a reason to weigh flexibility against higher utilisation.

WITH THE 38 PLANNED LOCATIONS

25%

more demand

37%

more same-hour shifting

DAILY MODE, REFERENCE CASE

	Optimisation	Greedy
Index gain	**0.373**	**0.327**
Time shift	**5.6 h**	**8.8 h**

WHERE THE VALUE IS LARGEST

Where room to receive, workload timing, and good wind and solar conditions meet in the same place and hour.

Source: Zhang and Zavala (2022) for the market design.

Limitations and next steps

Data and capacity

- Open map data can be incomplete. Missing floor counts default to one floor.
- IT share, power density and PUE are common or fleet-average values.
- Planned sites have approximate locations and estimated capacities.

Hourly inputs

- One load profile from UK data centres is applied to all locations.
- Availability measures wind and solar resources, not delivered renewable power or emissions.
- Only three historical weather years are used.

Model scope

- Latency, bandwidth, data sovereignty, hardware and transfer costs are not modelled.
- Total load after receiving work cannot exceed 95% of capacity.
- Projections scale today's network uniformly.

NEXT STEPS

1. Operator data on site capacity, hourly demand and job requirements.
2. Forecast-based scheduling with deadlines, arrival times and transfer limits.
3. Grid carbon intensity, curtailment and network constraints to value each shift.

Conclusion

SUB-QUESTION 1 – DEMAND

182 TWh/yr

is the estimated annual demand of the 87 clusters, including planned sites. By 2035, it reaches 259 to 562 TWh per year.

SUB-QUESTION 2 – SHIFTABILITY

4.18 TWh/yr

can move in daily mode and 2.52 TWh/yr in same-hour mode, 2.3% and 1.4% of demand. Room to receive is the limit.

SUB-QUESTION 3 – WIND AND SOLAR

10 times

higher wind and solar availability in the receive hours than in the send hours, in daily mode.

MAIN QUESTION

5.96–12.95 TWh/yr

can move by 2035 in daily mode. 47% of daily and 77% of same-hour shifting crosses world regions.

AI data centres can help with the energy flexibility problem. Their contribution depends more on spare room to receive work than on how flexible the work is.

Reference case: 85% utilisation, 25% flexible share, 6 send and receive hours per day, current and planned sites.

Thank you

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Questions and feedback are very welcome.

Code, data and all 16 notebooks are available at:
github.com/stefan-vasic/datacentre-demand-flexibility

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